

2.5 Clinical Overview

Silymarin 140 mg Hard Capsules

Generic application – Drug Administration of Vietnam (DAV)

Document type	Clinical Overview (abridged), ICH M4E(R2) / ACTD format
Product	Silymarin 140 mg hard capsules (hypothetical generic product)
Active substance	Standardised dry extract of milk thistle fruit (<i>Silybum marianum</i> (L.) Gaertn., fructus)
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List of Abbreviations

Abbreviation	Definition
ABCB1	ATP-binding cassette subfamily B member 1 (gene encoding P-glycoprotein)
ACTD	ASEAN Common Technical Dossier
ADR	Adverse drug reaction
ALT	Alanine aminotransferase
AUC	Area under the plasma concentration–time curve
CI	Confidence interval
C_{max}	Maximum plasma concentration
CTD	Common Technical Document
CYP	Cytochrome P450
DAV	Drug Administration of Vietnam
DI & ADR	Drug Information and Adverse Drug Reaction (National Centre)
DNPH	Dinitrophenylhydrazine (assay method)
EMA	European Medicines Agency
EU	European Union
f₂	Similarity factor (dissolution profile comparison)
HMPC	Committee on Herbal Medicinal Products
HPLC	High-performance liquid chromatography
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
NAFLD	Non-alcoholic fatty liver disease (now termed MASLD) ^a
NAS	NAFLD Activity Score
NASH	Non-alcoholic steatohepatitis (now termed MASH) ^a
NF-κB	Nuclear factor kappa B
Ph. Eur.	European Pharmacopoeia
RCT	Randomised controlled trial
RR	Relative risk
SyNCH	Silymarin in NASH and C Hepatitis (study programme)
t_½	Elimination half-life
TGF-β1	Transforming growth factor beta 1

^a The terms NAFLD and NASH are retained in this document as used in the cited trials. Current nomenclature: metabolic dysfunction-associated steatotic liver disease (MASLD) and metabolic dysfunction-associated steatohepatitis (MASH).

2.5 Clinical Overview

Key message This Clinical Overview supports a generic application for Silymarin 140 mg hard capsules. The benefit–risk balance is considered positive for supportive use in liver disorders, within the approved indication of the reference product and with the limitations of the efficacy evidence clearly acknowledged.

Table 1. Product and application summary

Item	Details
Product	Silymarin 140 mg hard capsules (hypothetical generic product)
Active substance	Standardised dry extract of milk thistle fruit (<i>Silybum marianum</i> (L.) Gaertn., fructus), equivalent to 140 mg silymarin
Application type	Generic application referencing a silymarin product already registered in Vietnam
Dossier format	ICH M4E(R2) / ACTD, Module 2.5 (abridged) [1]
Proposed posology	Adults: one capsule three times daily, swallowed whole with water

2.5.1 Product Development Rationale

Key message The product was developed as a quality-controlled generic alternative to an established silymarin medicine. No new clinical studies were conducted; the application relies on pharmaceutical equivalence to the reference product and on published literature.

Silymarin is a mixture of flavonolignans extracted from milk thistle fruit. Its principal components are silibinin (silybin A and B), isosilybin, silychristin and silydianin. Milk thistle preparations have a long history of use for hepatic and digestive complaints in Europe and are widely used as hepatoprotective agents in Vietnam.

The regulatory position on efficacy is cautious. In the 2018 EU herbal monograph, the EMA Committee on Herbal Medicinal Products (HMPC) accepted milk thistle preparations on the basis of traditional use only, namely the relief of digestive symptoms and the support of liver function once serious disease has been excluded by a physician. The HMPC reviewed the clinical studies in liver disease but was unable to draw firm conclusions owing to poor study design, small sample sizes and heterogeneous dosing [2].

Two principles therefore underpin this overview:

- **Quality determines equivalence.** For a herbal extract, the extraction solvent, drug–extract ratio and silymarin assay determine comparability with the reference product. The declared silymarin content depends on the analytical method: 108.2 mg determined by HPLC (Ph. Eur.) corresponds to 140 mg determined by the older DNPH method [3]. The 140 mg label claim must therefore specify the assay method.
- **Claims must be proportionate to the evidence.** The proposed indication follows the approved wording of the reference product, and the limitations of the efficacy data are stated explicitly (Section 2.5.4).

2.5.2 Overview of Biopharmaceutics

Key message Because silymarin is poorly soluble and marketed products differ widely in bioavailability, equivalence cannot be inferred from composition alone. It is supported by comparative dissolution testing and a comparative bioavailability study.

Silibinin, the principal and most pharmacologically active flavonolignan, is lipophilic and poorly water-soluble. Systemic exposure after oral administration is low for two reasons: rapid conjugation in the intestinal wall and liver, and efflux of the parent compound and its conjugates back into the intestinal lumen and into bile [4]. Commercial silymarin products differ substantially in silibinin content, dissolution and oral bioavailability [5].

A biowaiver is therefore not justified. The biopharmaceutic package supporting this application is summarised in Table 2.

Table 2. Biopharmaceutic studies supporting equivalence

Study	Design	Acceptance criterion
Comparative dissolution	Test vs reference product at pH 1.2, 4.5 and 6.8; 12 units per condition	Similarity factor $f_2 \geq 50$
Comparative bioavailability	Single-dose, two-way crossover study in healthy adults under fasting conditions	90% CI of the test/reference geometric mean ratio for C_{max} and AUC of silibinin within 80.00–125.00%

The analyte is total silibinin (silybin A + B), measured after enzymatic deconjugation, because unconjugated silibinin circulates in plasma at very low concentrations.

2.5.3 Overview of Clinical Pharmacology

Key message Silymarin acts on several hepatoprotective pathways in preclinical models, reaches peak plasma concentrations within a few hours and carries a low risk of clinically relevant drug interactions at therapeutic doses.

2.5.3.1 Pharmacodynamics

The proposed mechanisms of action are antioxidant (free-radical scavenging and preservation of glutathione), antifibrotic (inhibition of hepatic stellate cell activation and TGF- β 1 expression), anti-inflammatory (inhibition of NF- κ B) and toxin-blocking (reduced hepatocellular uptake of certain toxins). Silibinin may also promote hepatocyte regeneration by stimulating ribosomal RNA synthesis [6]. These effects have been demonstrated mainly in cell and animal models; no validated pharmacodynamic marker links them to clinical outcomes.

2.5.3.2 Pharmacokinetics

The disposition of silibinin after oral administration is illustrated in Figure 1. Rapid conjugation and enterohepatic recycling explain the low systemic exposure and the sensitivity of exposure to formulation differences (Section 2.5.2).

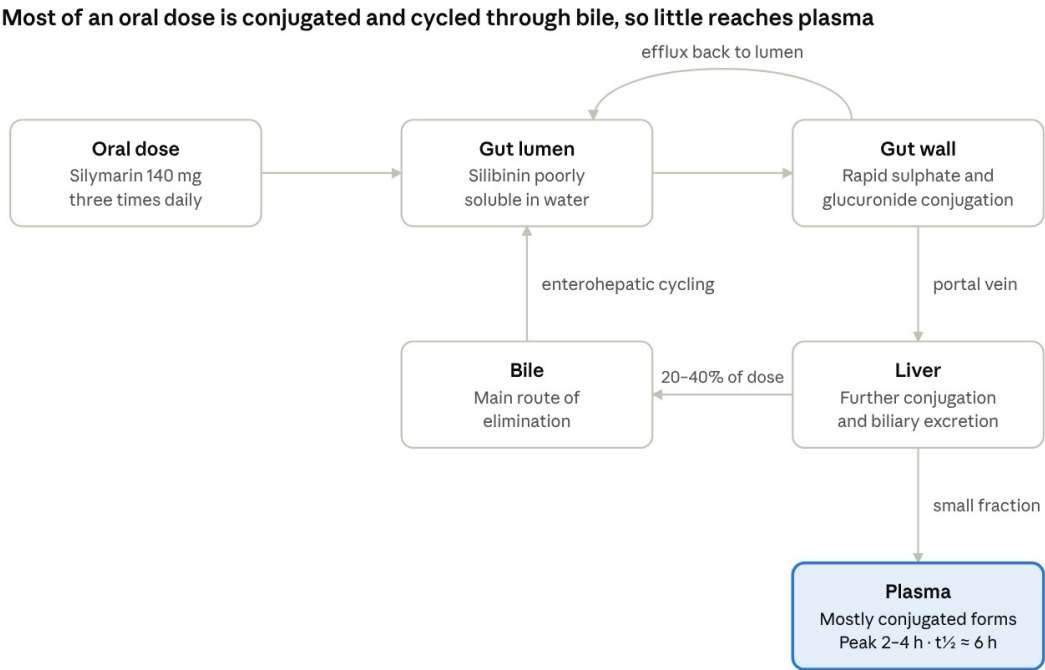


Figure 1. Disposition of silibinin after oral administration of silymarin

Sources: HMPc draft monograph [3]; Tvrdý et al. [4]; Tighe et al. [6].

Table 3. Key pharmacokinetic parameters of silibinin

Parameter	Silibinin (principal flavonolignan)	Reference
Time to peak plasma concentration	2–4 h	[6]
Plasma half-life (t½)	Approximately 6 h	[6]
Metabolism	Rapid sulphate and glucuronide conjugation in the intestinal wall and liver	[4]
Elimination	Predominantly biliary (approximately 20–40% of the dose), with enterohepatic recycling	[3]

Inter-individual variability in exposure is considerable. Hepatic impairment and transporter polymorphisms, such as ABCB1 C3435T, may alter silibinin disposition [6].

2.5.3.3 Drug Interactions

In human hepatocytes, silymarin produces negligible CYP inhibition even at supratherapeutic concentrations [6]. In vitro inhibition of metabolising enzymes and transporters has generally not translated into clinically relevant interactions in human studies [4]. As a precaution, caution is nevertheless advised with narrow-therapeutic-index drugs cleared by glucuronidation, such as sirolimus.

2.5.4 Overview of Efficacy

Key message No new efficacy studies were required for this generic application. The published evidence is inconsistent: one early trial in cirrhosis reported a survival benefit, but subsequent trials and meta-analyses have not confirmed an effect on mortality or disease progression.

2.5.4.1 Randomised Controlled Trials

The key randomised controlled trials of silymarin in chronic liver disease are summarised in Table 4, and their principal endpoint results are presented in Figure 2.

Table 4. Key randomised controlled trials of silymarin in liver disease

Study	Design and population	Silymarin regimen	Main findings
Ferenci et al., 1989 [7]	Double-blind RCT; 170 patients with alcoholic or non-alcoholic cirrhosis	140 mg three times daily vs placebo; mean follow-up 41 months	4-year survival 58% vs 39% (P = 0.036); in subgroup analyses, benefit confined to alcoholic cirrhosis and Child–Pugh class A
Parés et al., 1998 [8]	Multicentre double-blind RCT; 200 patients with alcoholic cirrhosis	150 mg three times daily vs placebo; 2 years	No effect on survival or clinical course
Fried et al., 2012 (SynCH) [9]	Multicentre double-blind RCT; 154 patients with chronic hepatitis C after failed interferon therapy	420 or 700 mg three times daily vs placebo; 24 weeks	No significant reduction in serum ALT (primary endpoint)
Wah Kheong et al., 2017 [10]	Double-blind RCT; 99 patients with biopsy-proven NASH	700 mg three times daily vs placebo; 48 weeks	Primary endpoint (≥30% reduction in NAS) not met; fibrosis improvement in 22.4% vs 6.0% (P = 0.023; secondary endpoint)
Navarro et al., 2019 (SynCH) [11]	Phase II multicentre double-blind RCT; 78 non-cirrhotic patients with NASH	420 or 700 mg three times daily vs placebo; 48–50 weeks	Primary endpoint (≥2-point NAS improvement) met in 19% (420 mg), 15% (700 mg) and 12% (placebo); P = 0.79

Only the 1989 cirrhosis trial met its primary endpoint

Patients reaching each endpoint (%), silymarin vs placebo

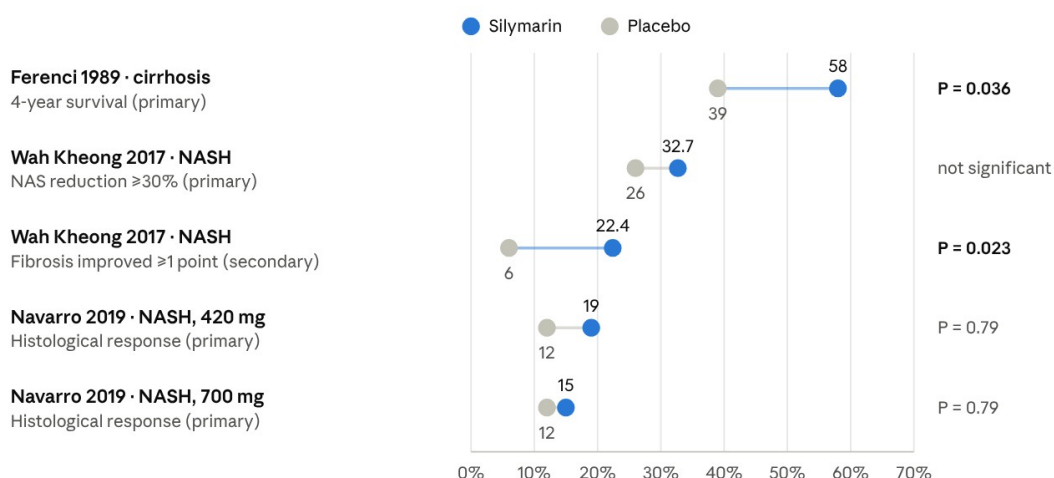


Figure 2. Endpoint results of key randomised trials

Sources: Ferenci et al. [7]; Wah Kheong et al. [10]; Navarro et al. [11]. Parés et al. [8] and Fried et al. [9] reported no between-group difference and are not plotted.

The only statistically significant primary result derives from the oldest trial. The fibrosis benefit observed in 2017 was a secondary endpoint and was not reproduced in the 2019 NASH trial.

2.5.4.2 Meta-analysis

A Cochrane systematic review of 18 randomised trials (1,088 patients) with alcoholic and/or hepatitis B or C virus liver disease found no significant effect of milk thistle on all-cause mortality (RR 0.78; 95% CI 0.53–1.15), complications of liver disease (RR 0.95; 95% CI 0.83–1.09) or liver histology. The authors emphasised the lack of high-quality evidence [12].

2.5.4.3 Interpretation for this Application

- The proposed dose of 140 mg three times daily corresponds to the regimen of the only trial reporting a survival benefit [7].
- The evidence for clinically meaningful outcomes is not robust, and trial results vary with formulation, dose and population. Efficacy claims in the product information should therefore remain within the approved wording of the reference product.
- The fibrosis signal in NASH is hypothesis-generating only and does not support a new indication.

2.5.5 Overview of Safety

Key message Silymarin has a well-characterised and favourable safety profile. The principal risks are mild gastrointestinal effects and rare allergic reactions in individuals sensitive to plants of the Asteraceae family.

2.5.5.1 Adverse Reactions

The EU herbal monograph lists mild gastrointestinal symptoms (dry mouth, nausea, epigastric discomfort, gastric irritation and diarrhoea), headache, and allergic reactions including skin rash,

pruritus, urticaria, asthma and anaphylaxis; the frequency of these reactions is not known [2]. In the randomised trials summarised in Section 2.5.4, doses of up to 700 mg three times daily were well tolerated, and the Cochrane review found no significant increase in adverse events compared with placebo [12].

2.5.5.2 Nonclinical Safety

Silymarin showed no toxicity in rats and mice after single oral doses of 2,500–5,000 mg/kg, nor in rats receiving up to 2,500 mg/kg/day for 12 months. Adequate genotoxicity, carcinogenicity and reproductive toxicity studies have not been performed [3].

2.5.5.3 Special Populations and Proposed Product Information

Table 5. Proposed safety wording for the product information

Topic	Proposed position	Basis
Hypersensitivity	Contraindicated in patients with hypersensitivity to silymarin or to other plants of the Asteraceae family	EU herbal monograph [2]
Paediatric population (< 18 years)	Not recommended owing to the absence of adequate safety and efficacy data	EU herbal monograph [2]
Pregnancy and breastfeeding	Not recommended; safety has not been established	EU herbal monograph [2]
Drug interactions	No clinically relevant interactions established; caution with narrow-therapeutic-index drugs cleared by glucuronidation	Section 2.5.3.3
Overdose	No cases reported	EU herbal monograph [2]

2.5.5.4 Post-marketing Surveillance

The applicant will monitor safety through routine pharmacovigilance and will report adverse drug reactions to the National Centre of Drug Information and Adverse Drug Reaction Monitoring (National DI & ADR Centre) in accordance with DAV requirements.

2.5.6 Benefits and Risks Conclusions

Key message The benefit–risk balance of Silymarin 140 mg hard capsules is considered positive for the approved indication of the reference product, provided that equivalence is confirmed by the comparative bioavailability study and that efficacy claims in the product information are not extended.

2.5.6.1 Benefits

Silymarin has plausible hepatoprotective mechanisms and decades of clinical use. One randomised trial at the proposed dose reported improved survival in patients with cirrhosis [7]. However, subsequent trials and a Cochrane meta-analysis did not confirm effects on mortality, complications or histology [8–12]. The expected benefit is therefore supportive and modest.

2.5.6.2 Risks

Adverse reactions are predominantly mild and gastrointestinal. Allergic reactions are rare but may be serious in individuals sensitive to Asteraceae plants. The risk of clinically relevant drug interactions is low.

2.5.6.3 Key Uncertainties and Risk Management

Table 6. Key uncertainties and risk management

Uncertainty	Management
Bioavailability differs between silymarin products	Comparative dissolution and bioavailability study against the reference product (Section 2.5.2)
Limited high-quality efficacy evidence	Indication restricted to the approved wording of the reference product; no new claims
Risk of delayed diagnosis of serious liver disease	Package leaflet advises patients to consult a physician if symptoms persist or worsen
No data in children or in pregnancy and breastfeeding	Use not recommended in these populations

2.5.6.4 Overall Conclusion

On the basis of the data presented, the applicant considers that the benefit–risk balance is positive and that Silymarin 140 mg hard capsules are approvable for the approved indication of the reference product.

2.5.7 Literature References

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